

NICHOLAS ATACHA

ASSIGNMENT 1.3: SECURITY STRATEGY DEVELOPMENT – 3 – YEAR SECURITY ROADMAP

ORGANIZATION: FUTURE BANK

FOCUS: YEAR 1 SECURITY ROADMAP

DATE: 4TH AUGUST 2026

Executive Summary

Future Bank is growing its digital banking services and needs a strong security foundation. The main threats facing the bank are phishing attacks, ransomware, data leaks, insider threats and penalties from CBN and NDPA for non-compliance.

This Year 1 plan focuses on putting the right policies, people, and technology in place to manage risk. We will focus on 5 areas: Governance, Security Operations, Infrastructure, Applications, and Data. The plan is broken into 4 quarters to make it practical and achievable while supporting the bank's growth and protecting customer information.

Year 1 Security Roadmap

Area	What We Will Do	When	Cost vs Benefit
Governance, Risk & Compliance	Implement ISO 27001 and CBN Cybersecurity Framework. Create security policies, risk register, and vendor risk process. Appoint a CISO. Do quarterly internal audits.	Q1 – Q4 Q1: Assessment Q2-Q3: Policy Launch Q4: Audit	Cost: High. Consultants, GRC tools, CISO. Benefit: Meets regulatory requirements. Avoids fines. Improves management oversight. Value: High

Security Operations	Set up a 24/7 SOC. Deploy SIEM and EDR tools. Recruit and train SOC staff. Build and test an Incident Response plan.	Q2 – Q4 Q2: Buy Tools Q3: Build SOC Q4: Go Live 24/7	Cost: High. Tool licenses and staff salaries. Benefit: Detects attacks early. Reduces damage and downtime. Value: Very High
Infrastructure Security	Separate critical systems using network segmentation. Upgrade firewalls. Force MFA for all staff and vendors. Scan for vulnerabilities every and patch.	Q1 – Q3 Q1: MFA + Firewall Q2: Segmentation Q3: Vulnerability program	Cost: Medium. Hardware and software. Benefit: Blocks attackers from moving inside the network. Protects core banking systems. Value: High
Application Security	Review the code of Mobile and internet Banking apps for security issues. Deploy a web Application Firewall. Do penetration tests every quarter and fix issues.	Q2 – Q4 Q2: Code Review Q3: Deploy WAF Q4: Pen testing	Cost: Medium. WAF and pen test services. Benefit: Prevents fraud and theft of customer data. Protects the bank's reputation. Value: High
Data Security	Classify sensitive data. Encrypt data stored and data moving on the network. Use DLP to stop data from leaving. Set up encrypted backups and test disaster recovery.	Q1 – Q4 Q1-Q2: Classify + Encrypt Q3: DLP Q4: Backup/DR Test	Cost: Medium. Encryption, DLP, backup storage. Benefit: Complies with NDPA. Prevents data loss. Ensures we can recover. Value: Very High

Quarter by Quarter Plan

Q1: Build the Foundation

Set up governance, hire CISO, enforce MFA, upgrade firewall, and start classifying data.

Q2: Gain Visibility

Deploy SIEM, begin application code reviews, start network segmentation, and hire SOC team.

Q3: Strengthen Defenses

Train SOC team, deploy WAF and DLP, run vulnerability scans, and do first penetration test.

Q4: Operate and Review

Start 24/7 SOC monitoring, conduct internal audit, test backup and DR, and plan for Year 2.

How This Supports Future Bank

This plan is practical for a mid-sized bank. It tackles the biggest risks first:

Compliance and data protection. It also builds the team and tools needed to detect and respond to attacks. By following this roadmap, Future Bank will reduce risk, meet regulations, and keep customer trust.

Final Note

This Year 1 strategy gives Future Bank a clear path to improve security. It covers people, process, and technology. Executing this will put the bank in a strong position for Years 2 and 3.